

The method for using the flash_download_tool to flash the firmware generated by ESP-IDF to the ESP32 is as follows:

Step 1: Prepare firmware file

1. After compiling a VSCode + ESP-IDF project, the firmware file is generated by default in the project's **build** directory.
2. The following key files are required:

Bootloader file: **bootloader/bootloader.bin**

Partition table file: **partition_table/partition-table.bin**

Main application file: **app.bin** (located in the **build** directory of the project root directory). The generated **app.bin** file will have a different name depending on the project configuration.

Step 2: Download the Flash download tool

1. Download the tool from Espressif's official website:

Flash Download Tool Download Link:

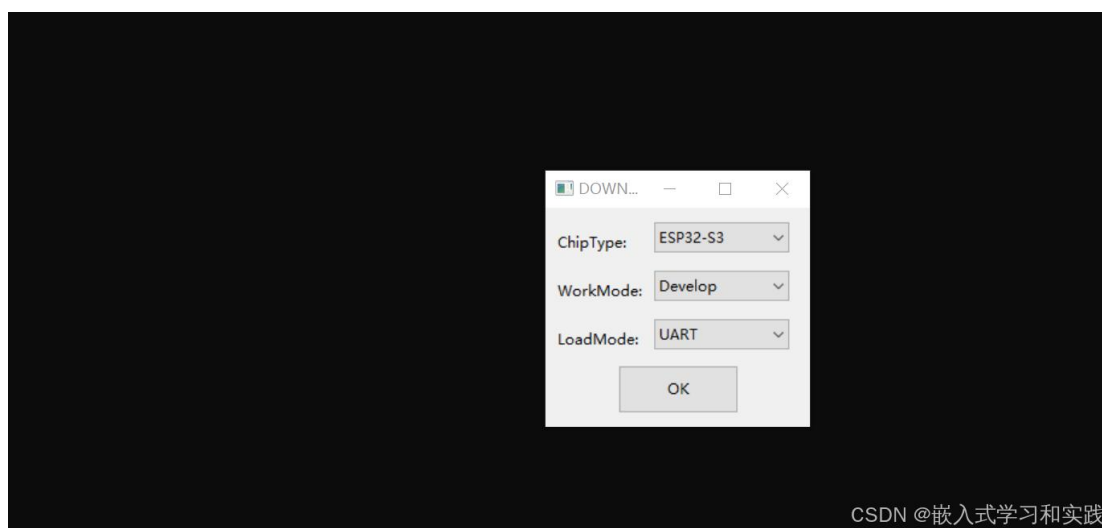
[https://www.espressif.com.cn/zh-hans/support/download/other-tools`](https://www.espressif.com.cn/zh-hans/support/download/other-tools)

Step 3: Configure the Flash download tool

1. Open the tool:

*Run the extracted "**flash_download_tool_x.x.x.exe**."

*Select the chip type (e.g., **ESP32** or **ESP32-S2/S3**).



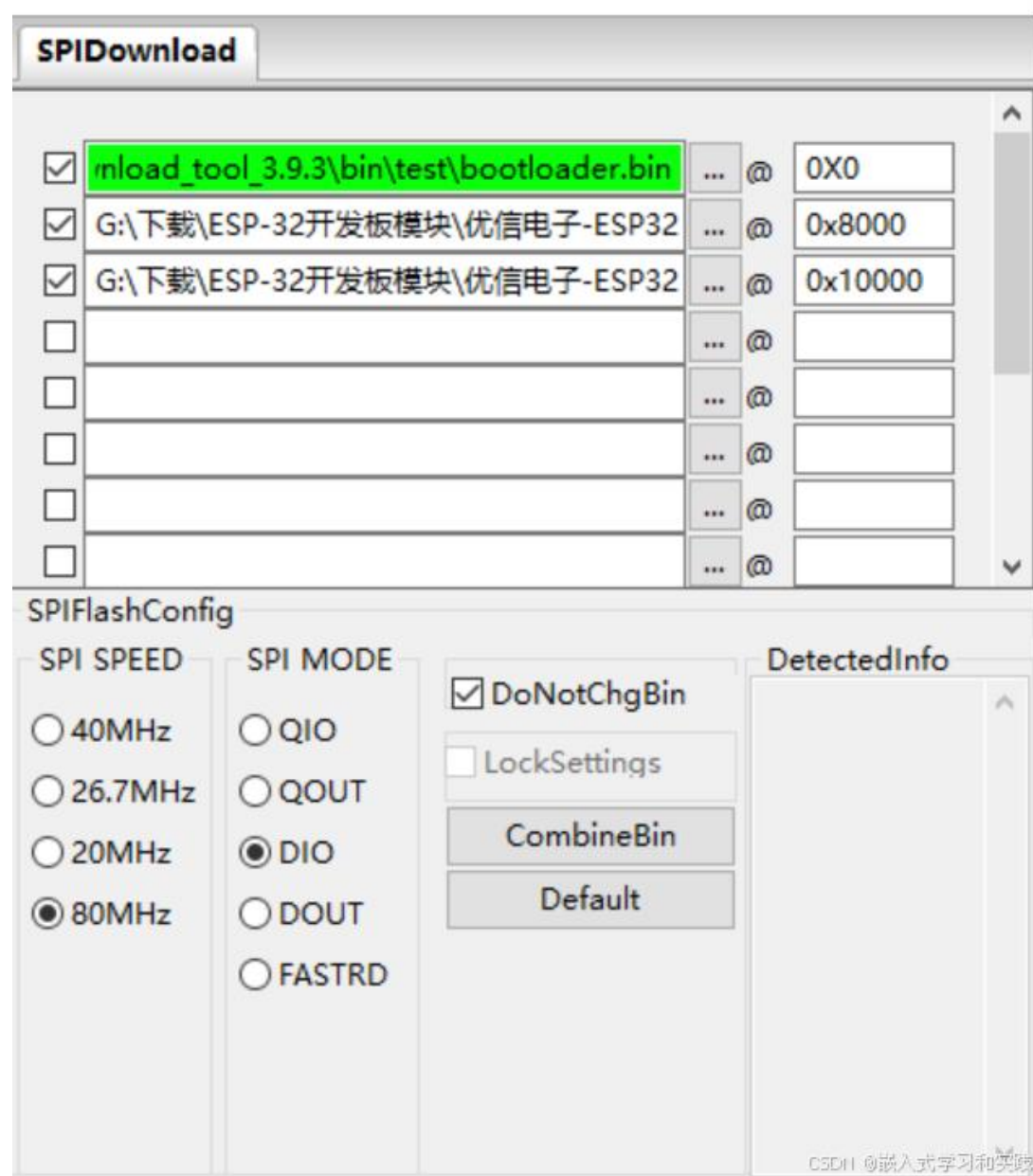
2. Configure flashing parameters

*Development Mode: Select **Develop** (default).

*Work Mode: Select **UART** (programming via serial port).

*SPI SPEED: Usually select **80MHz**.

*SPI MODE: Usually select **DIO or QIO** (consistent with project configuration).

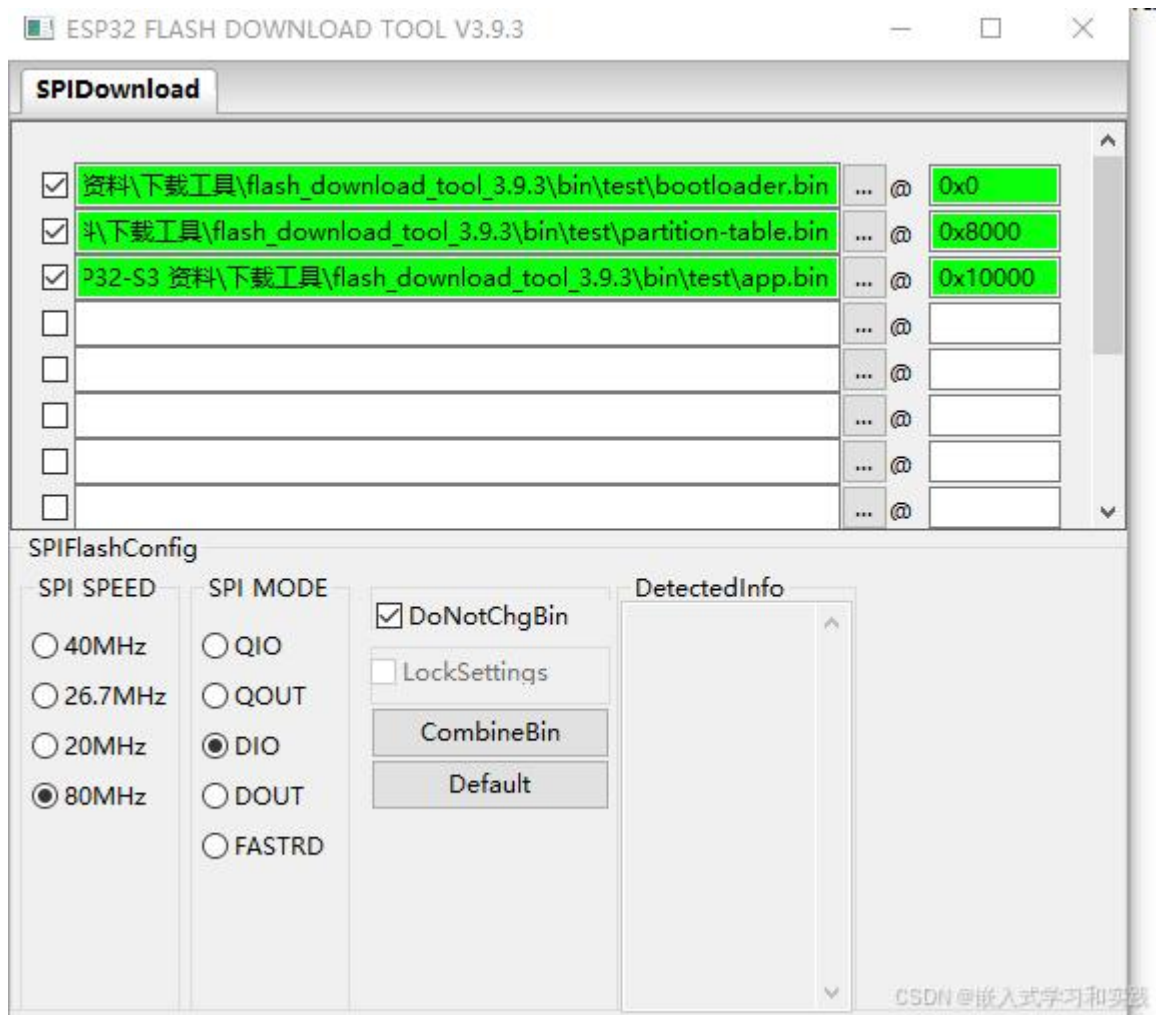


3. Add firmware file

Click the "... " button on the right, add the following files in sequence, and specify the burning address:

file path	Burning address (Hex)
bootloader.bin	0x0
partition-table.bin	0x8000
your_project.bin	0x10000

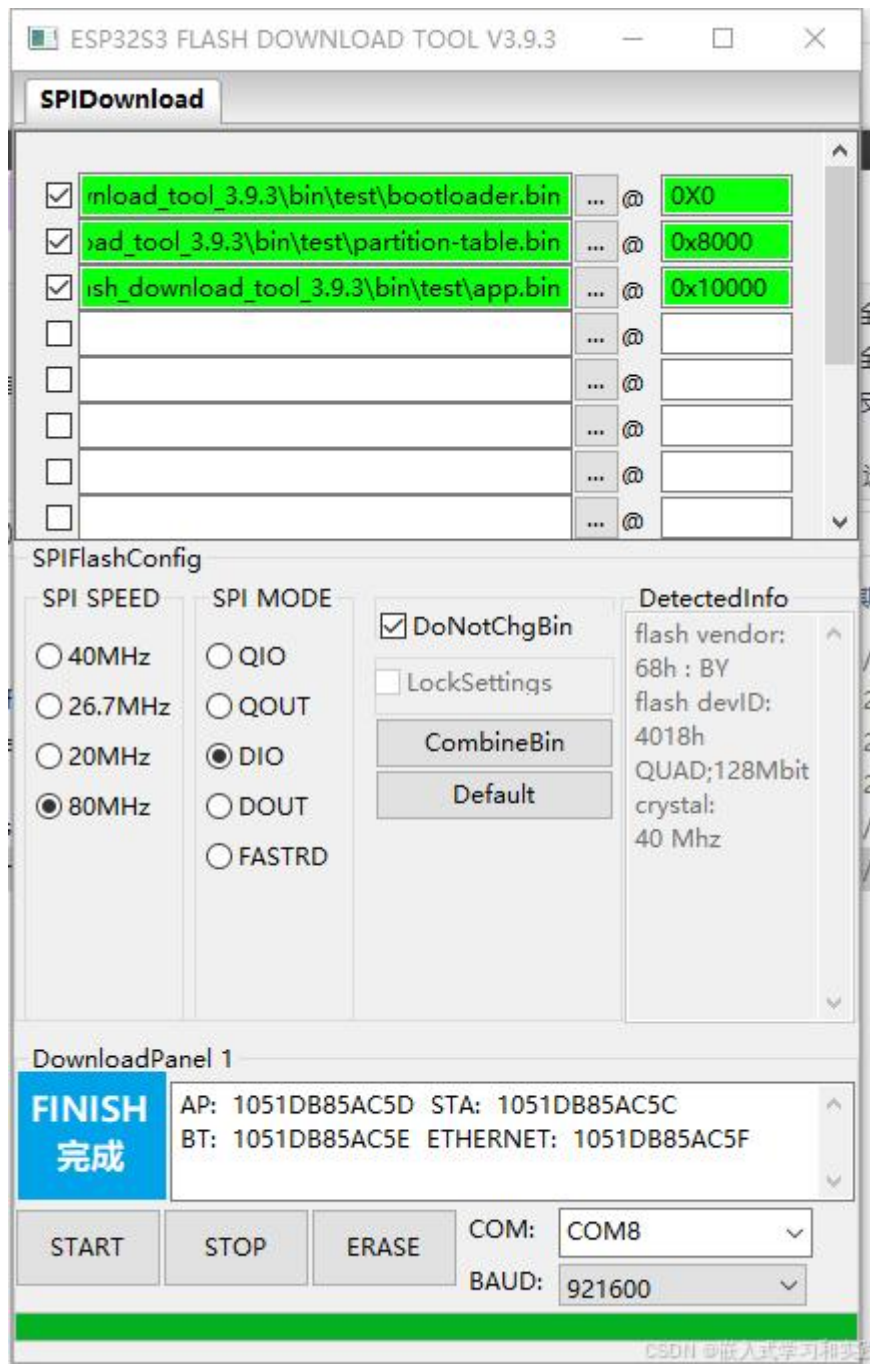
Note: The address may vary depending on the project configuration (such as a custom partition table).



4. Set serial port parameters

*COM Port: Select the serial port number for the ESP32 connection (e.g., COM8).

*Baudrate: 921600 is recommended (can be reduced to 115200 if download fails).



Step 4: Start flashing

1. Connecting the ESP32:

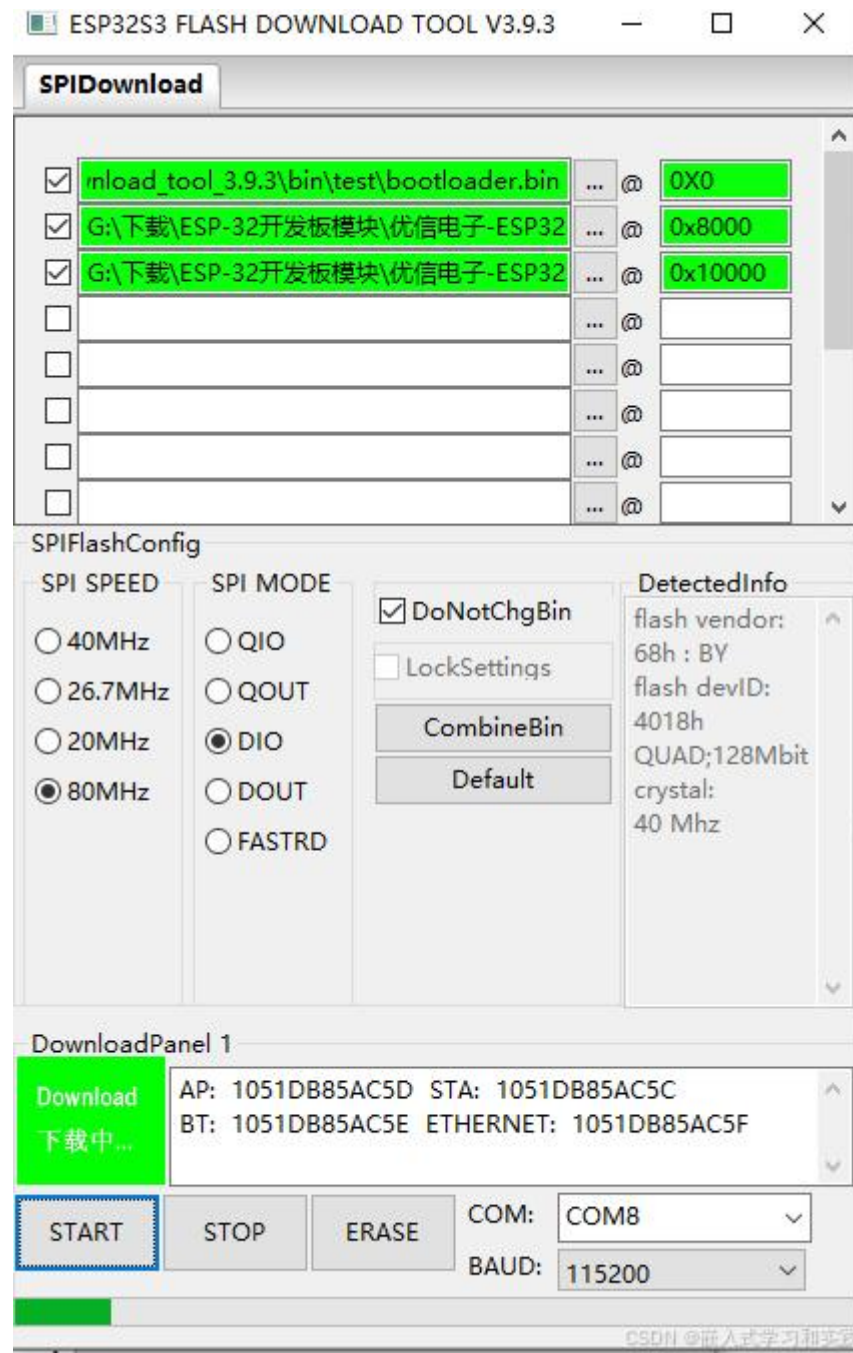
*Connect the ESP32 to your computer via USB, ensuring the serial driver is installed (e.g., CP210x or CH340).

*Ensure the development board is in download mode:

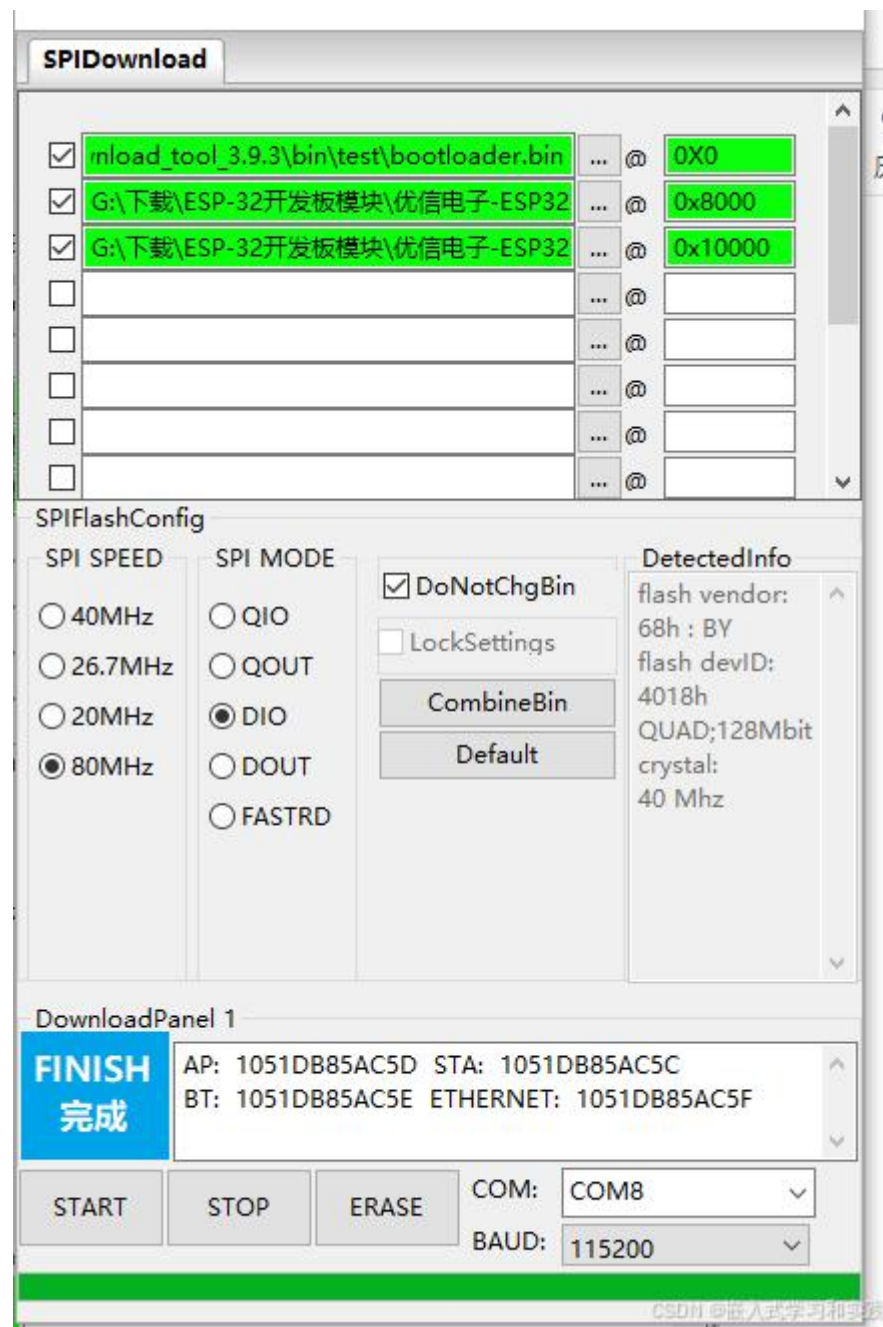
Press and hold the **BOOT** button → Briefly press the **RESET** button → Release **RESET** → Release **BOOT**.

2. Start the programming process:

*Click the **START** button at the bottom of the tool.



*Wait for the progress bar to complete; once it displays "FINISH," the burning process is successful.



Step 5: Verify firmware

Press the **RESET** button to restart the ESP32-S3.

Use a serial port tool (such as **idf.py monitor** or **PuTTY**) to check the serial port output and confirm that the program is running normally.

Common problems:

*COM port not found: Check if the USB driver is installed, or try a different USB cable.

*Burning failed: Lower the baud rate (e.g., 115200) or check the hardware connections.

*Address error: Verify that the address of partition-table.bin matches the project configuration.

By following the steps above, you can successfully burn the firmware generated by ESP-IDF into the ESP32-S3 device.

Reference - VSCode download process log based on ESP-IDF

```
1 正在执行任务: C:\Users\esp32\.espressif\python_env\idf5.2_py3.11_env\Scripts\python.exe C:\Users\esp32\esp\v5.2.5\esp
2  --before default_reset
3  --after hard_reset
4  --chip esp32s3 write_flash
5  --flash_mode dio
6  --flash_freq 80m
7  --flash_size 16MB
8  0x0 bootloader/bootloader.bin
9  0x10000 app.bin
10 0x8000 partition_table/partition-table.bin
11
12 esptool.py v4.8.1
13 Serial port COM4
14 Connecting...
15 Chip is ESP32-S3 (QFN56) (revision v0.2)
16 Features: WiFi, BLE, Embedded PSRAM 8MB (AP_3v3)
17 Crystal is 40MHz
18 MAC: 10:51:db:85:ac:5c
19 Uploading stub...
20 Running stub...
21 Stub running...
22 Changing baud rate to 460800
23 Changed.
24 Configuring flash size...
25 Flash will be erased from 0x00000000 to 0x00005fff...
26 Flash will be erased from 0x00010000 to 0x0001ffff...
27 Flash will be erased from 0x00080000 to 0x0008ffff...
28 SHA digest in image updated
29 Compressed 21632 bytes to 13321...
30 Wrote 21632 bytes (13321 compressed) at 0x00000000 in 0.7 seconds (effective 263.5 kbit/s)...
31 Hash of data verified.
32 Compressed 922880 bytes to 569390...
33 Wrote 922880 bytes (569390 compressed) at 0x00010000 in 14.4 seconds (effective 512.0 kbit/s)...
34 Hash of data verified.
35 Compressed 3072 bytes to 103...
36 Wrote 3072 bytes (103 compressed) at 0x00080000 in 0.1 seconds (effective 320.5 kbit/s)...
37 Hash of data verified.
38
39 Leaving...
40 Hard resetting via RTS pin...
```